



Cognitive Neuroscience in the Foundation Model Era: Lessons in Reproducibility

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brainhack-vandy.github.io

My 10 years of Brainhack

2016 - First Brainhack in Paris

2017 - First Brainhack Global

2019 - Neurohackademy

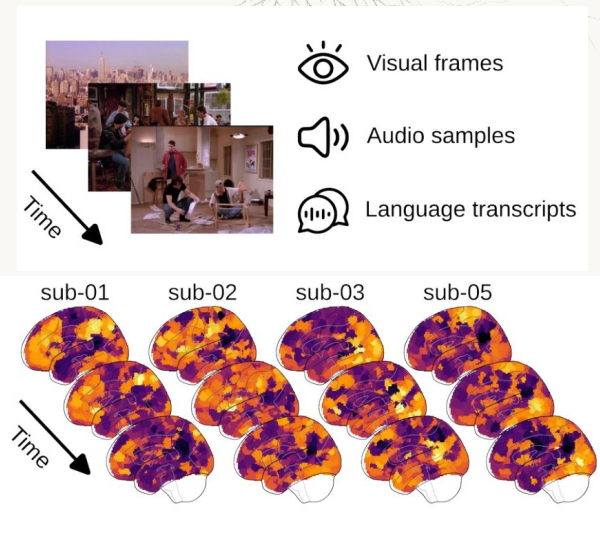
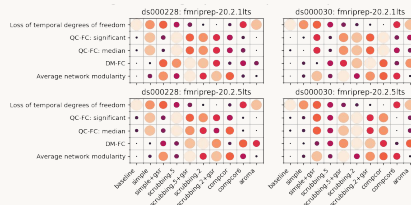
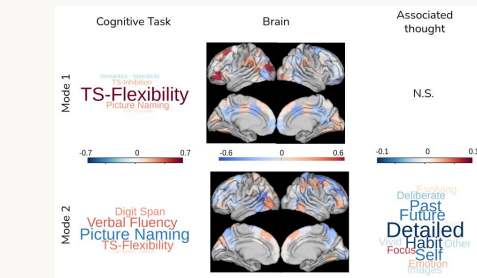
2021 to Present - Nilearn core developer

2022 - Cohost OHBM Brainhack Glasgow

2026 - First invited talk at Brainhack



What do I do in neuroscience?



PhD

University of York, UK

Postdoc

CRIUGM, Canada

Currently

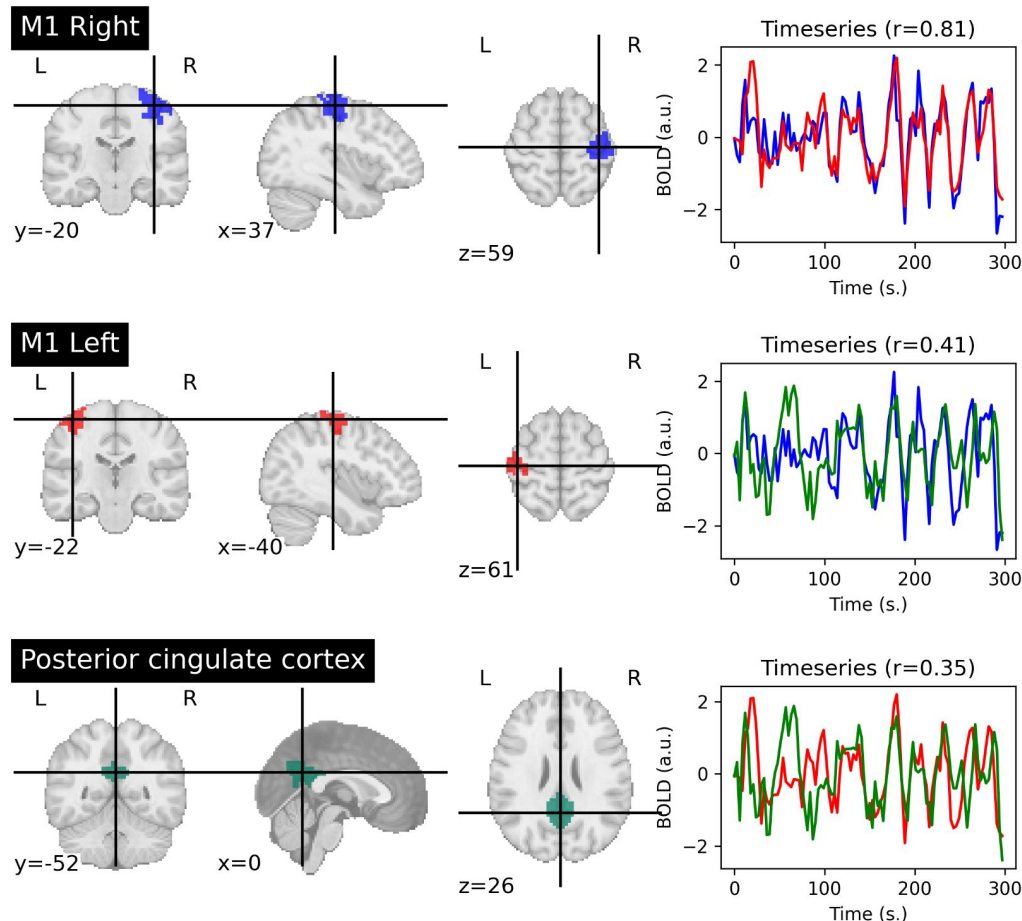
University of British Columbia,
Canada

Can brain activity predict psychiatric / mental health conditions?

Functional connectivity captures spontaneous brain activity in resting state fMRI

Figure generated by L Bellec using Python code and the Nilearn library, under CC-BY license

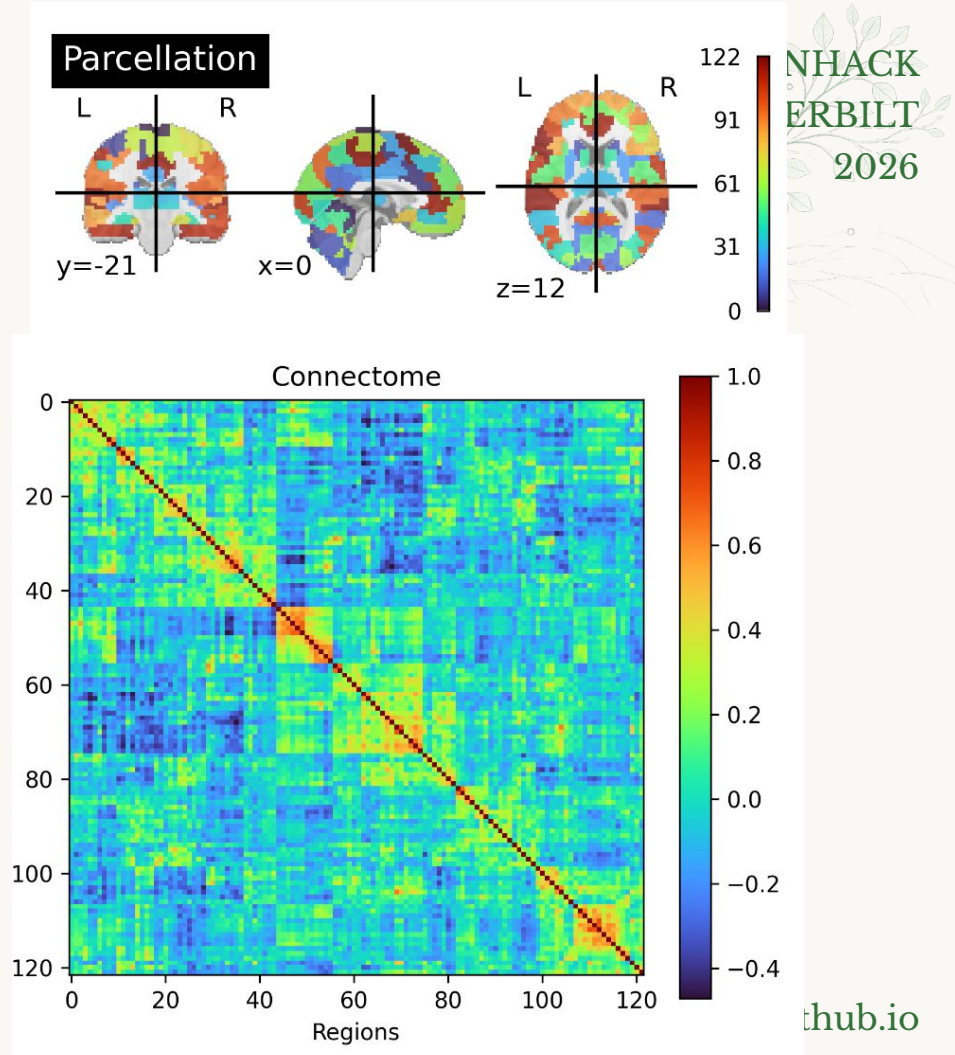
<https://methodes-cogneuro.github.io/connectivite.html>



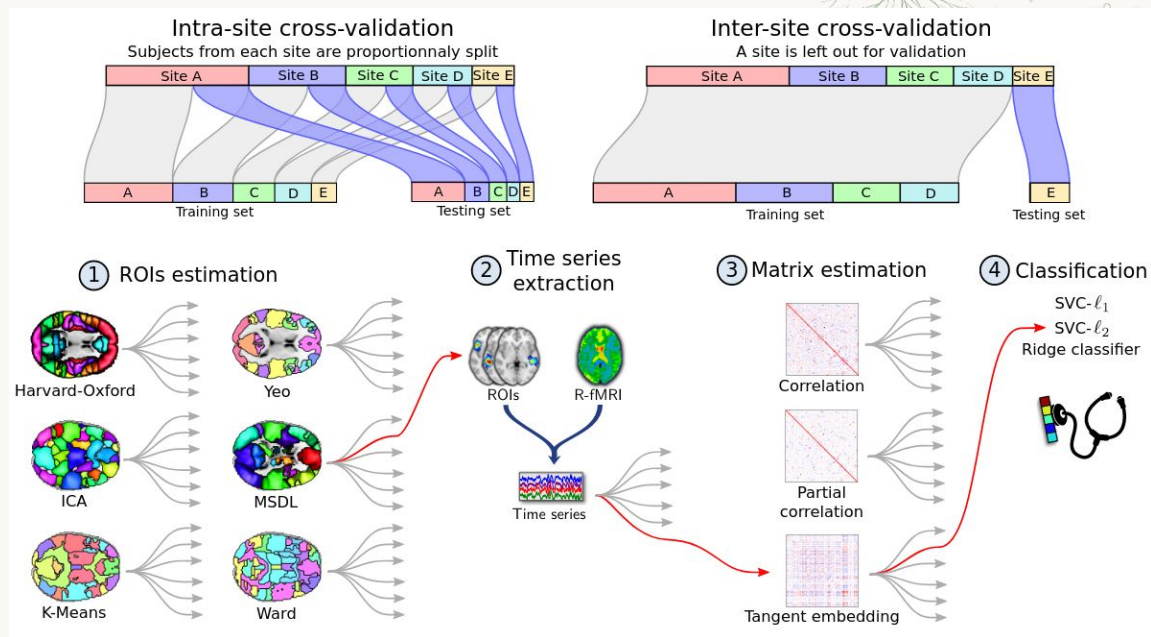
Functional connectivity captures spontaneous brain activity in resting state fMRI

Figure generated by L Bellec using Python code and the Nilearn library, under CC-BY license

<https://methodes-cogneuro.github.io/connectivite.html>



A classic workflow of diagnosis prediction with functional connectivity

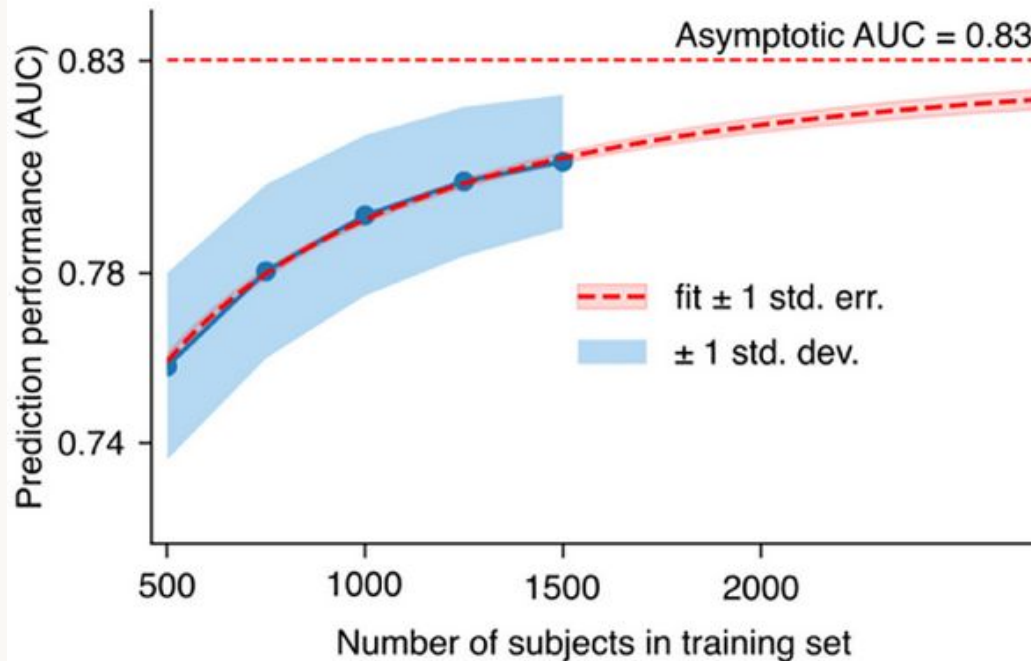


ABIDE dataset

Abraham et al. (2017) NeuroImage

Brain-based
biomarkers require
large sample size
for trait-level
prediction

Prediction for various sample sizes



Traut et al. (2022) NeuroImage

Reliably preprocess neuroimaging data and compare results

Wang et al. (2024)

PLOS Computational Biology

Preprocessing fMRI
data from 40,000
subjects with the
power of



Adolescent Brain Cognitive Development™



BRAINHACK
VANDY
2026

```
dicomdir/  
├── 1208200617178_22/  
│   ├── 1208200617178_22_8973.dcm  
│   ├── 1208200617178_22_8943.dcm  
│   ├── 1208200617178_22_2973.dcm  
│   ├── 1208200617178_22_8923.dcm  
│   ├── 1208200617178_22_4473.dcm  
│   ├── 1208200617178_22_8783.dcm  
│   ├── 1208200617178_22_7328.dcm  
│   ├── 1208200617178_22_9264.dcm  
│   ├── 1208200617178_22_9967.dcm  
│   ├── 1208200617178_22_3894.dcm  
│   └── 1208200617178_22_3899.dcm  
├── 1208200617178_23/  
├── 1208200617178_24/  
└── 1208200617178_25/
```



```
my_dataset/  
├── participants.tsv  
├── sub-01/  
│   ├── anat/  
│   │   └── sub-01_T1w.nii.gz  
│   ├── func/  
│   │   ├── sub-01_task-rest_bold.nii.gz  
│   │   └── sub-01_task-rest_bold.json  
│   └── dwi/  
│       ├── sub-01_dwi.nii.gz  
│       ├── sub-01_dwi.json  
│       ├── sub-01_dwi.bval  
│       └── sub-01_dwi.bvec  
├── sub-02/  
├── sub-03/  
└── sub-04/
```

Gap in the workflow for generating data for model training

MRI data

```

my_dataset/
├── participants.tsv
├── sub-01/
│   ├── anat/
│   │   └── sub-01_T1w.nii.gz
│   ├── func/
│   │   ├── sub-01_task-rest_bold.nii.gz
│   │   └── sub-01_task-rest_bold.json
│   └── dwi/
│       ├── sub-01_dwi.nii.gz
│       ├── sub-01_dwi.json
│       ├── sub-01_dwi.bval
│       └── sub-01_dwi.bvec
├── sub-02/
├── sub-03/
└── sub-04/
    
```



Preprocessing



Feature extraction

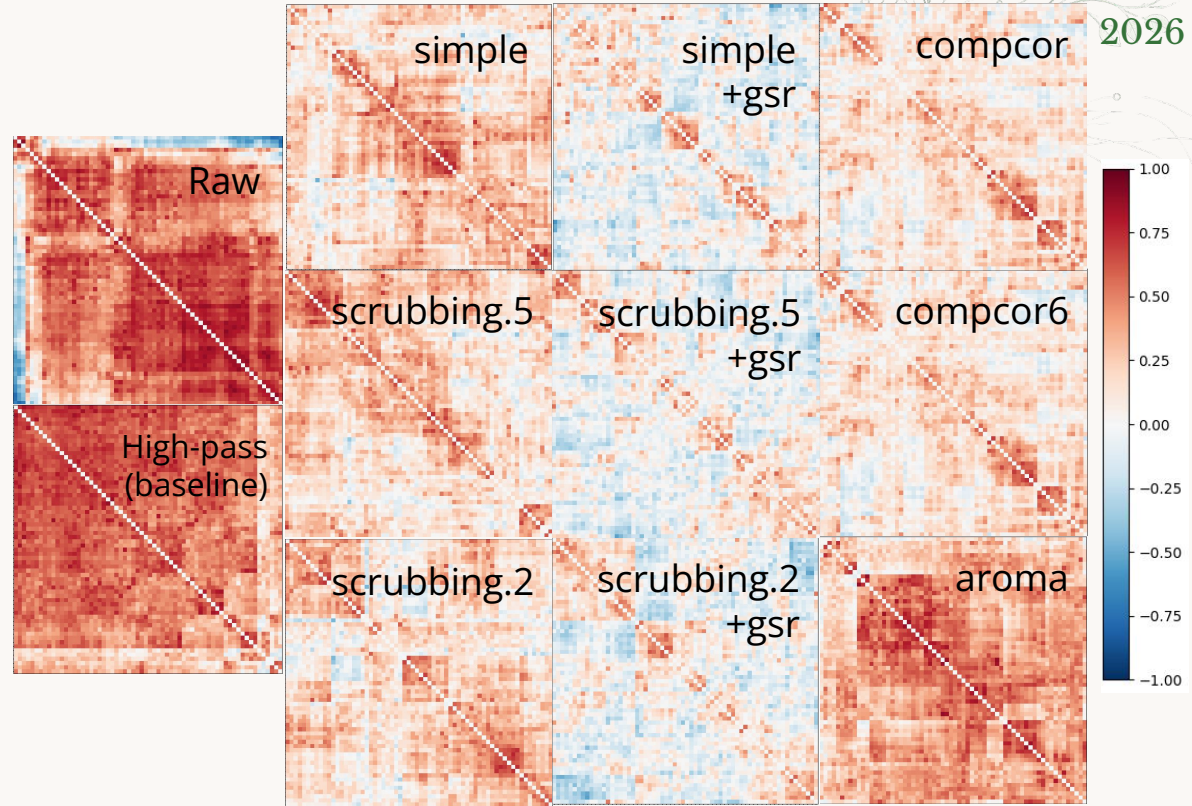


Denoising?

Denoising is complex

An overview of denoised connectomes from all possible and methodologically-sound combinations of fMRIPrep provided confound regressors.

Wang et al. (2024)
PLOS Computational Biology



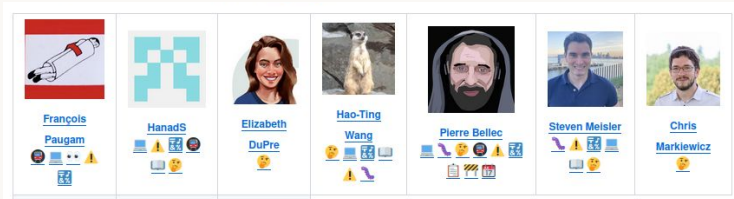
load_confounds:

Community-driven
project that
streamlines
denoising



```
from nilearn.interfaces.fmriprep import \
    load_confounds_strategy

confounds, sample_mask = \
    load_confounds_strategy(
        fmri_filenames,
        denoise_strategy="simple"
    )
```



[brainhack-vandy.github.io](https://github.com/brainhack/vandy)

Preprocessed 40,000 subjects



MRI data

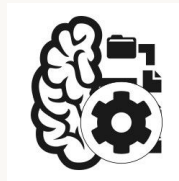
```
my_dataset/  
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│   ├── func/  
│   │   ├── sub-01_task-rest_bold.nii.gz  
│   │   └── sub-01_task-rest_bold.json  
│   └── dwi/  
│       ├── sub-01_dwi.nii.gz  
│       ├── sub-01_dwi.json  
│       ├── sub-01_dwi.bval  
│       └── sub-01_dwi.bvec  
├── sub-02/  
├── sub-03/  
└── sub-04/
```

Preprocessing



giga-connectome

giga-auto-qc



Wang et al. (2025) JOSS

Feature extraction



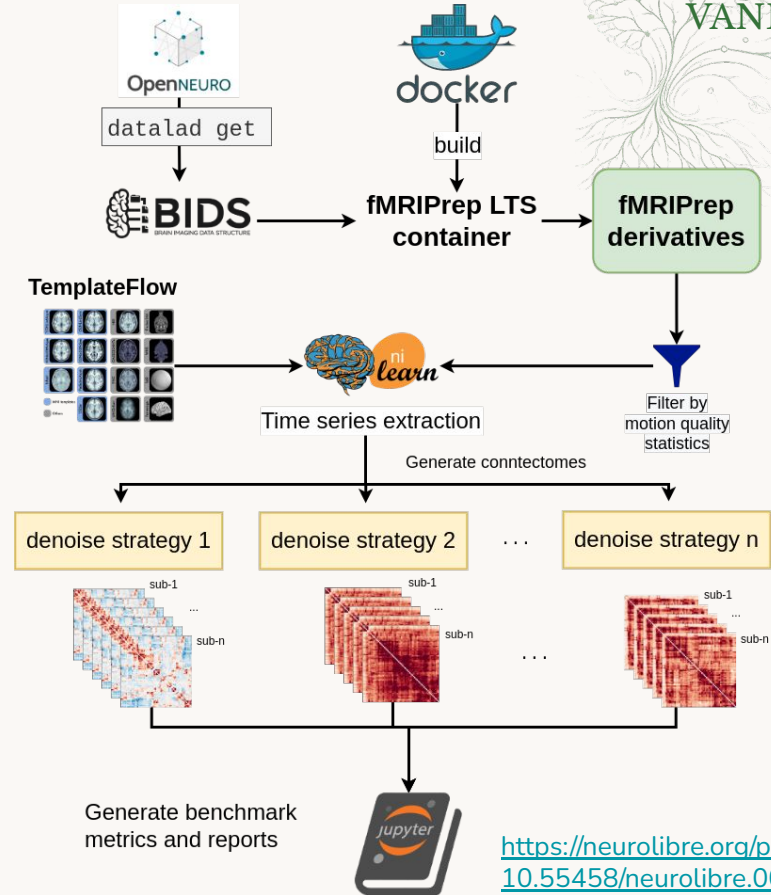
Denoising guideline built with reproducibility in mind

**2023 WINNERS of The Neuro
Irv and Helga Cooper Foundation
Open Science Prizes**



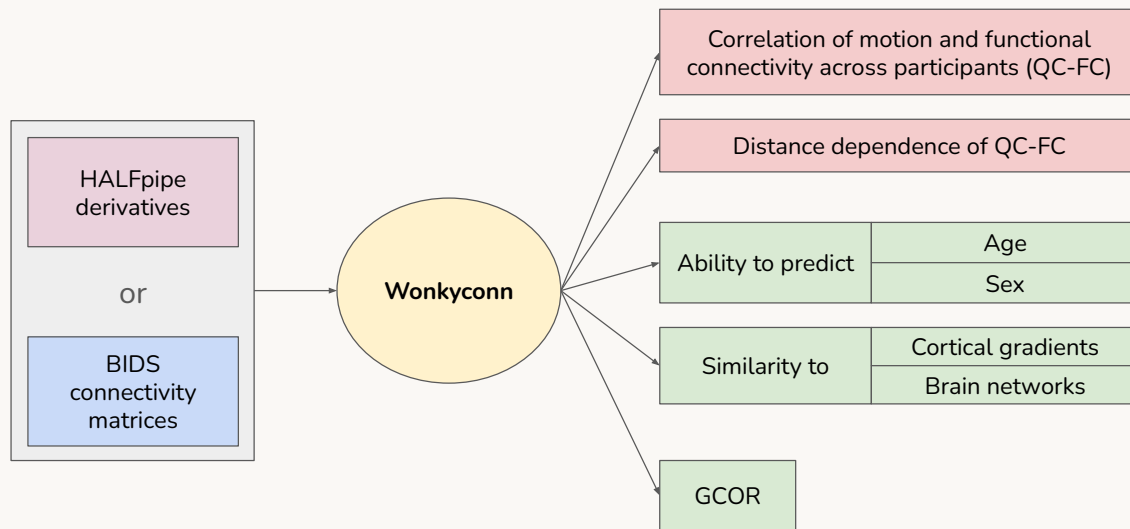
Hao-Ting Wang
Continuous Evaluation of Denoising
Strategies in Resting State fMRI
Connectivity Using fMRIPrep and
Nilearn, OULAM, Montreal, Canada

Wang et al. (2024)
PLOS Computational Biology



<https://neurolibre.org/papers/10.55458/neurolibre.00012>
[brainhack-vandy.github.io](https://github.com/brainhack-vandy)

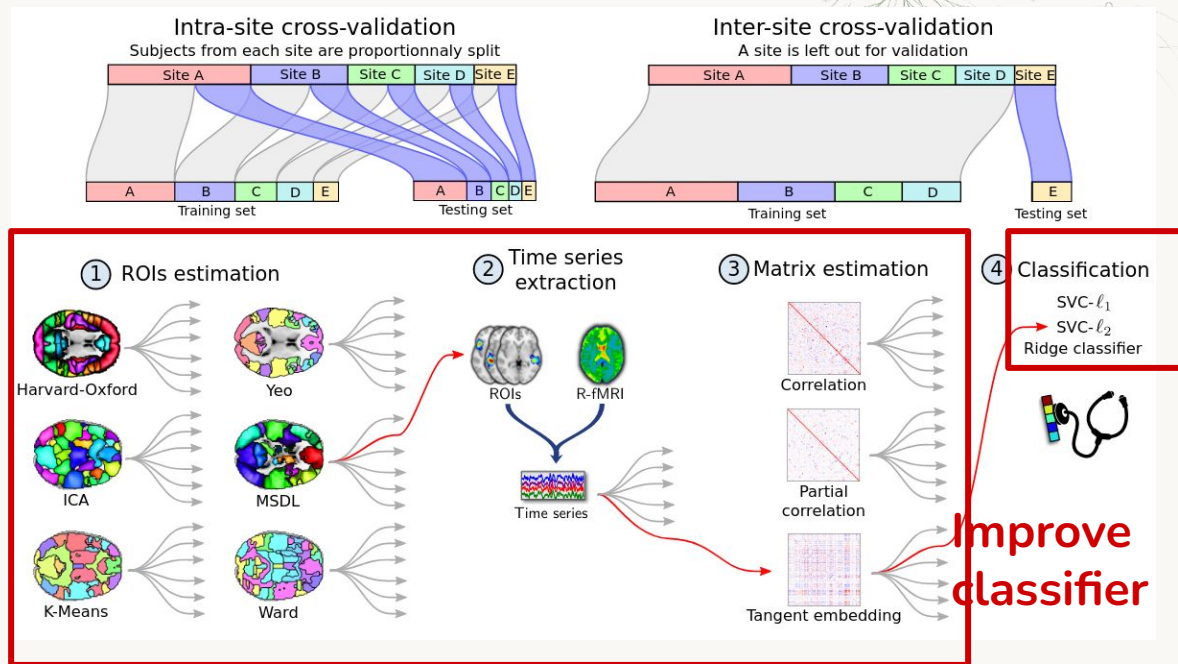
Integrating denoising benchmark in ENIGMA resting state fMRI workflow



Current gaps in applications of fMRI foundation models

Wang et al. (under preparation)

A classic workflow of brain-based biomarkers on multi-site clinical sample



ABIDE dataset

Abraham et al. (2017) NeuroImage

Improve features

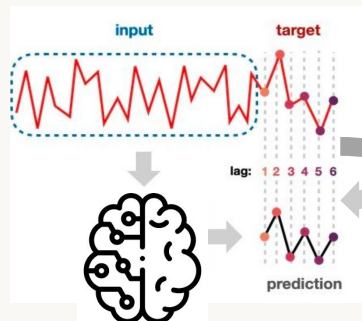
Improve classifier

Utility of foundation models

Pretraining



A lot of data



Evaluation

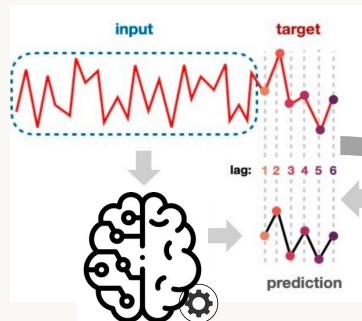
Downstream tasks



Fine-tuning



Specialised dataset

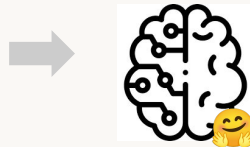


Evaluation

Downstream tasks



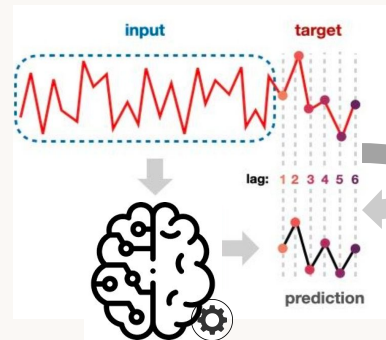
Fine-tuning: Can changes in model predictions across time be used as markers for clinical progression?



BrainLM (Caro, 2024)
BrainJEPA (Dong, 2024)



Quality Control



Evaluation



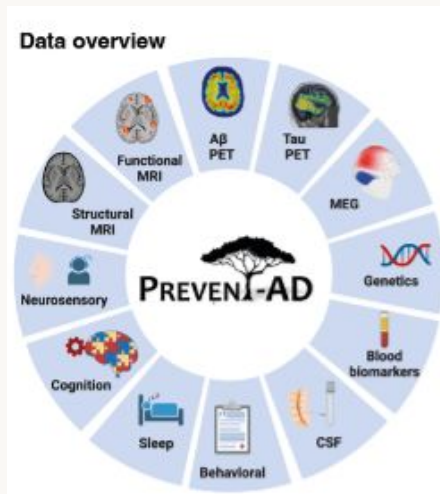
Downstream tasks



Target dataset:



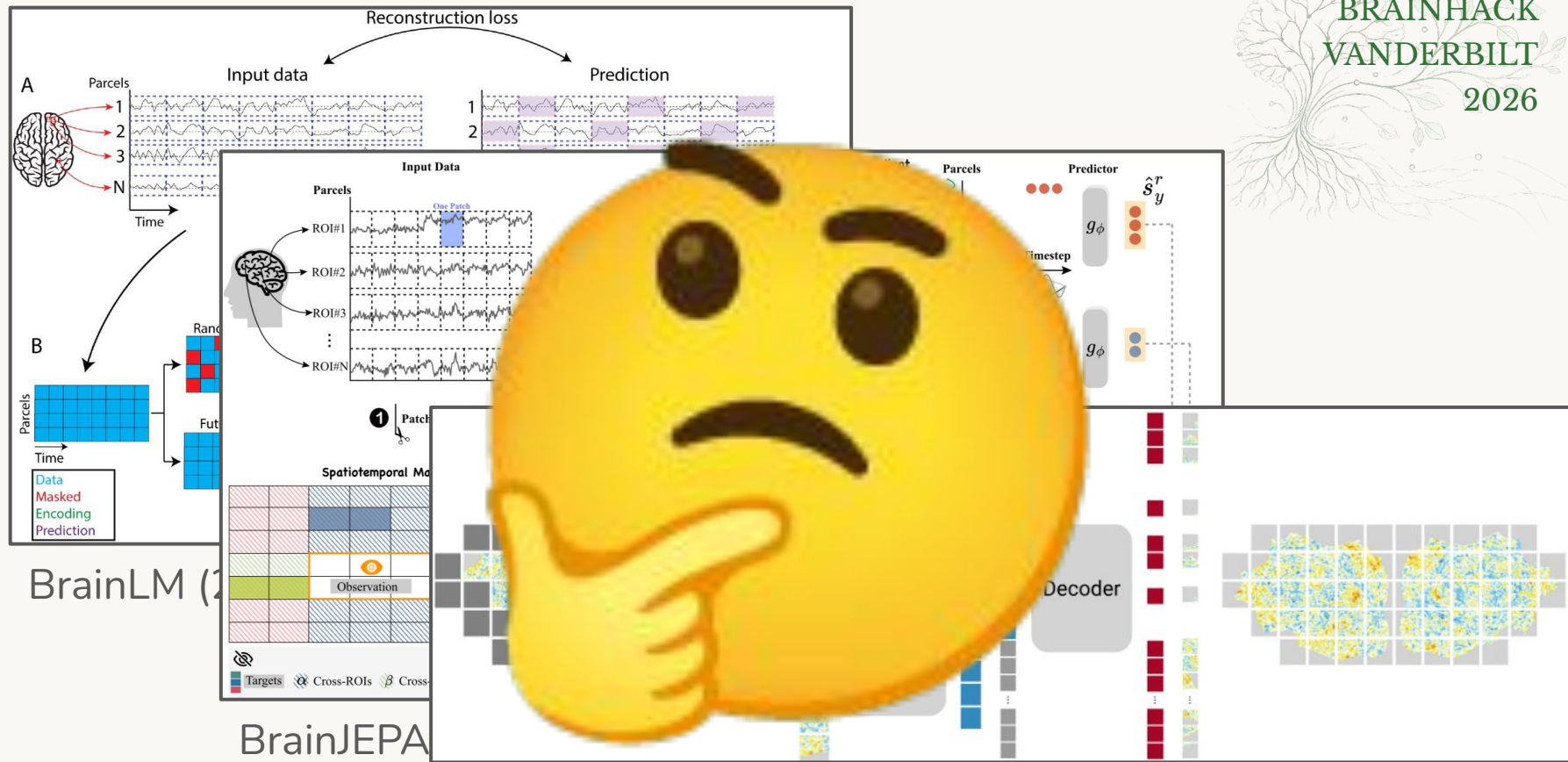
- Cognitively unimpaired older adults with a first-degree family history of Alzheimer's Disease
- Baseline data collection before any clinical symptoms
- Longitudinal tracking



Prediction Targets:

- Demographic information:
Is there meaningful signal present?
 - Sex
 - Age - split half
- Clinical targets:
Is there clinically relevant information?
 - Prognosis: MCI status
 - Diagnosis: Amyloid SUVR of grey matter relative to the cerebellum, (cut-off = 1.26; N=218)

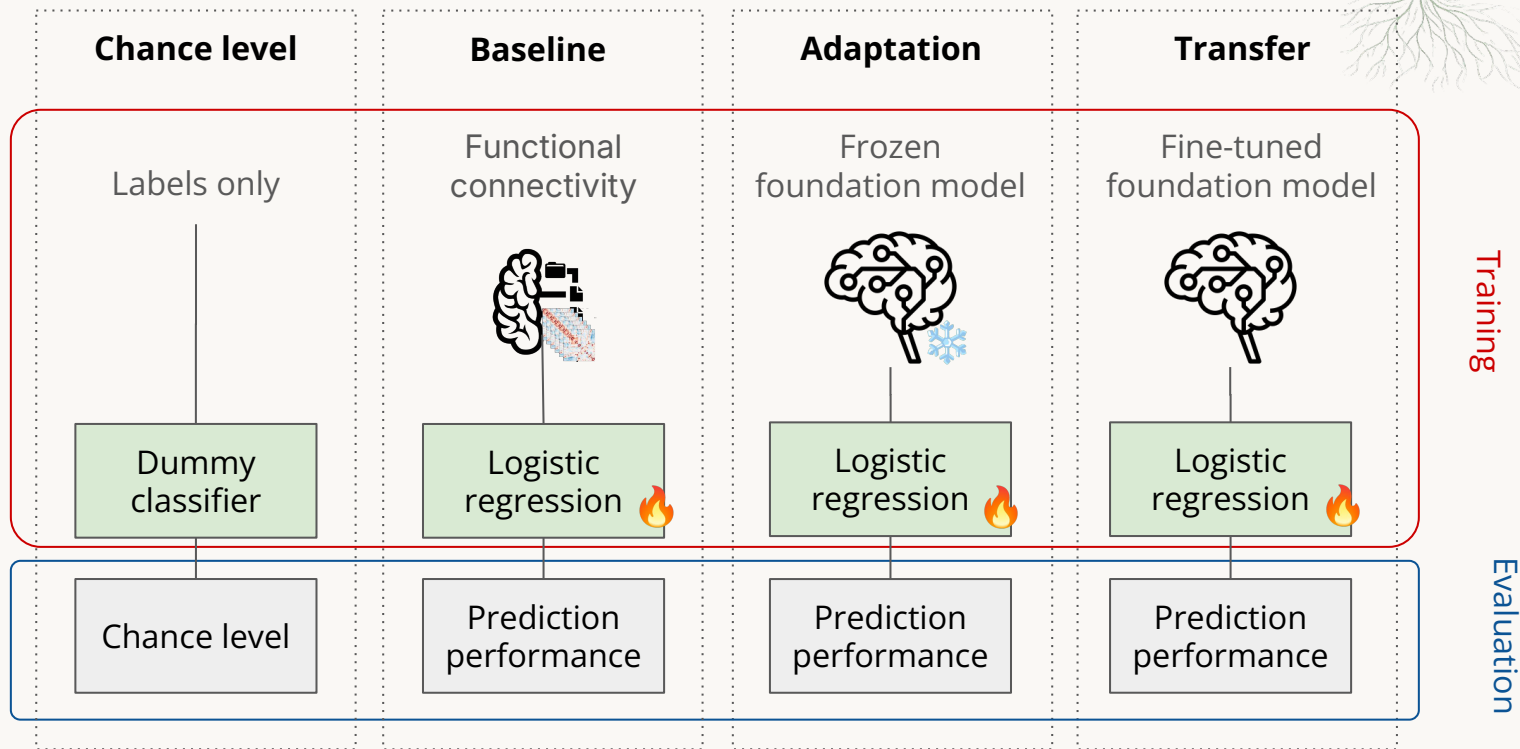




Objectives

- Identify the challenges an independent group of researchers would face when deploying foundation models on a new dataset.
- Compare the foundation models with a common baseline in the fMRI machine learning literature.
- Examine whether foundation models improve performance in downstream applications.

Experiment setup

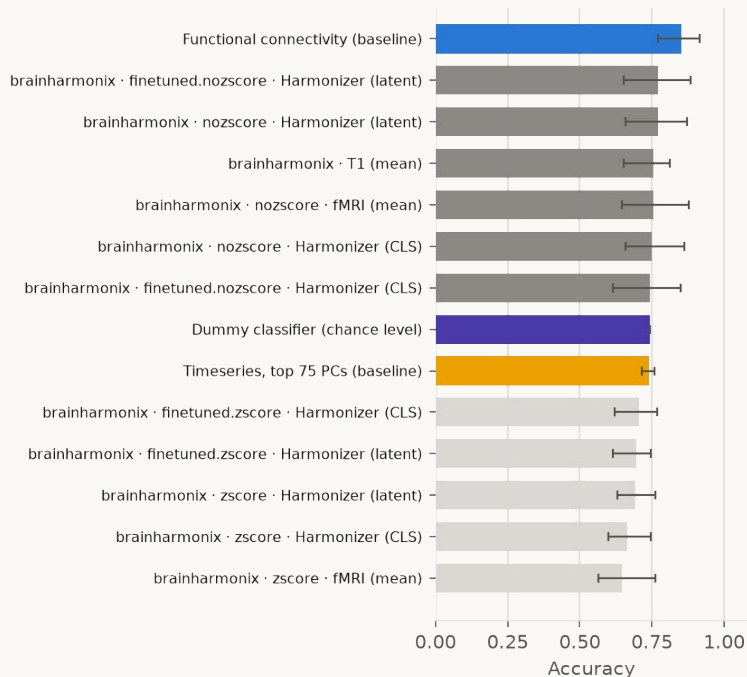


Reusing code is difficult

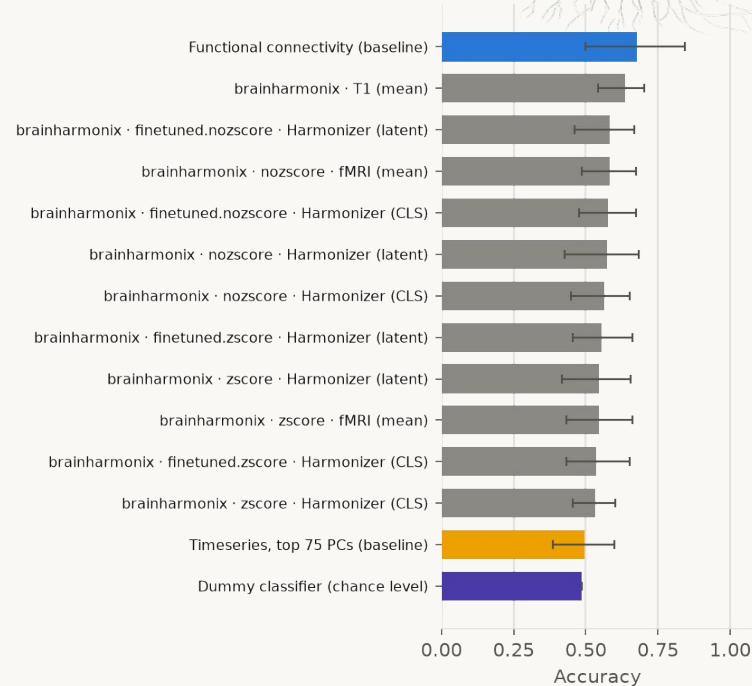
	BrainLM				BrainHarmonix	
Preprocessing						
fMRI voxel level post processing	Denosing: high-pass filter, 6 motion parameters + WM + CSF + global signal					
	Within subject normalization: z-score	No normalization			Within subject normalization: z-score	No normalization
Atlas	A424				Schaefer 400	
Time series extraction	BrainLM workflow	Nilearn workflow	BrainLM workflow	Nilearn workflow	Nilearn workflow	Nilearn workflow
Global normalization**	BrainLM workflow	Robust scalar	BrainLM workflow	Robust scalar	Robust scalar	Robust scalar
Workflow matching neuroimaging research standard (giga-connectome workflow) Closest to original workflow based on released code Outputs for generating benchmark baseline **Only applied for foundation model training / fine-tuning						

Sex and age prediction to verify the presence of signal

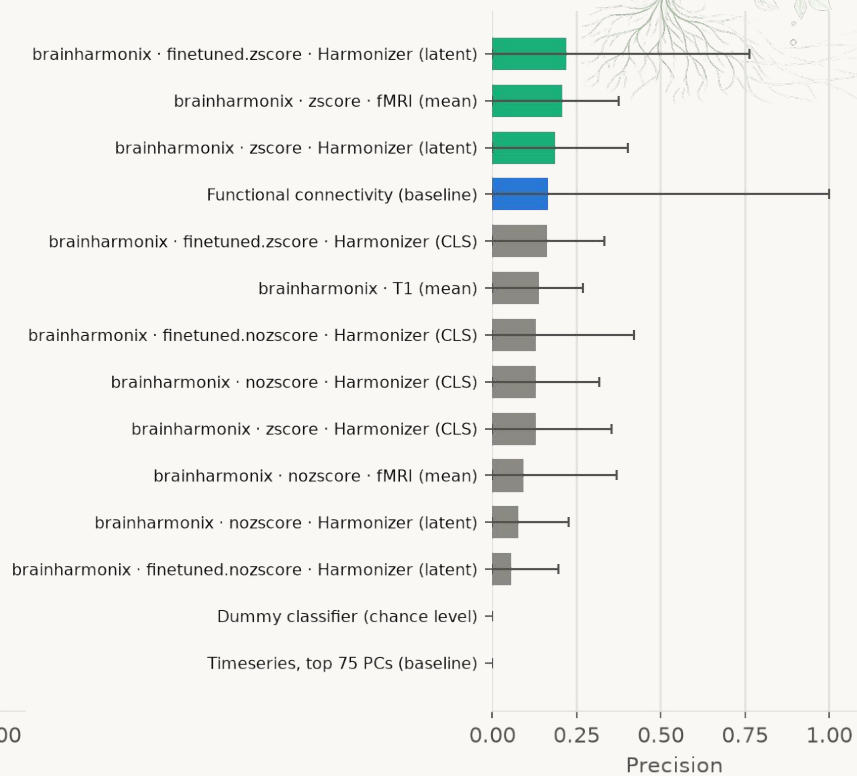
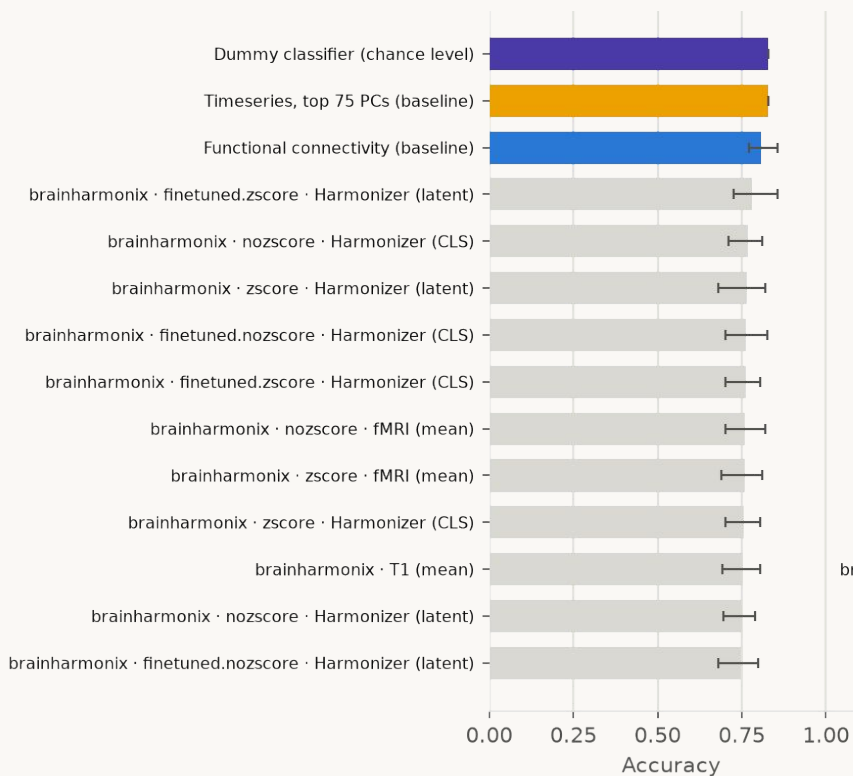
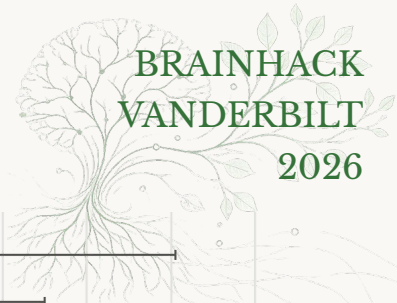
Sex



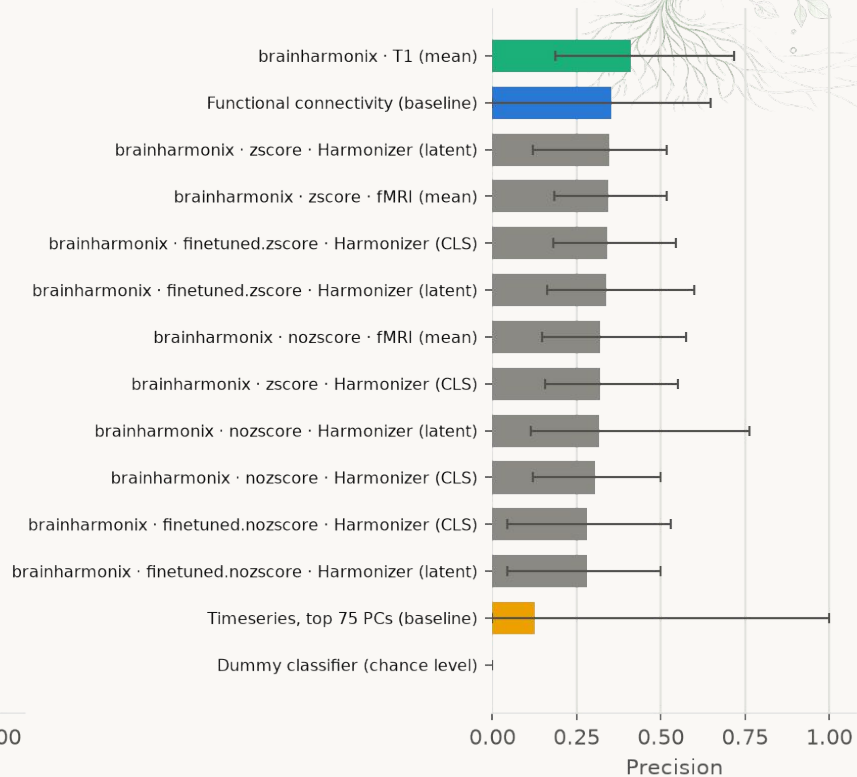
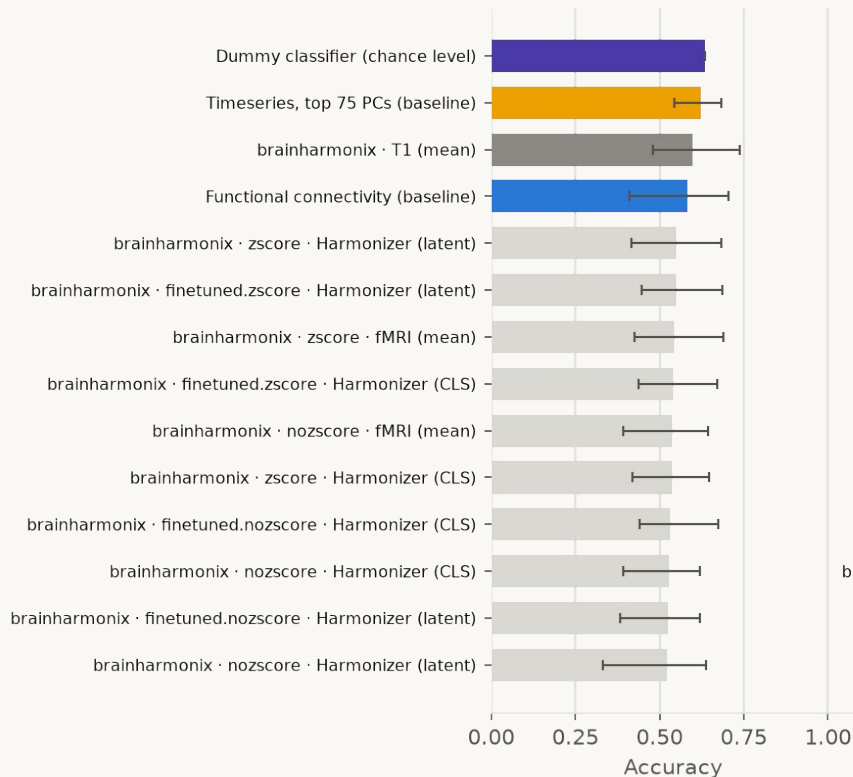
Age (median split)



MCI status



Amyloid SUVR > 1.26



Summary

Downstream transfer seen in large language models cannot yet be assumed for fMRI foundation models.

1. Independent application of foundation models was technically challenging
2. Functional connectivity with regularized logistic regression is an important baseline, difficult to beat
3. Limited performance of foundation models on downstream tasks
4. Precision (true positive rates/ recall rate) is an important metric for clinical prediction targets

Imaging foundation model through a lens of cognitive neuroscience

Dedicated to the memory of Jonny Smallwood

Lessons learned from surveying the fMRI “foundation models”

What I do well:

fMRI data processing ✨

Efficient data loading ✨

Model training procedure and tools ✨

Framework for downstream prediction ✨

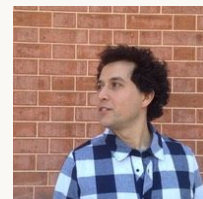
What the field needs:

fMRI data processing grounded in understanding of the data ⚠️

Meaningful and common benchmark beyond sex and age prediction ⚠️

Reproducible code ⚠️

Data privacy aware model ⚠️



It's not all doom and gloom: nanoGPT provides directions for large model training

available GPT implementations

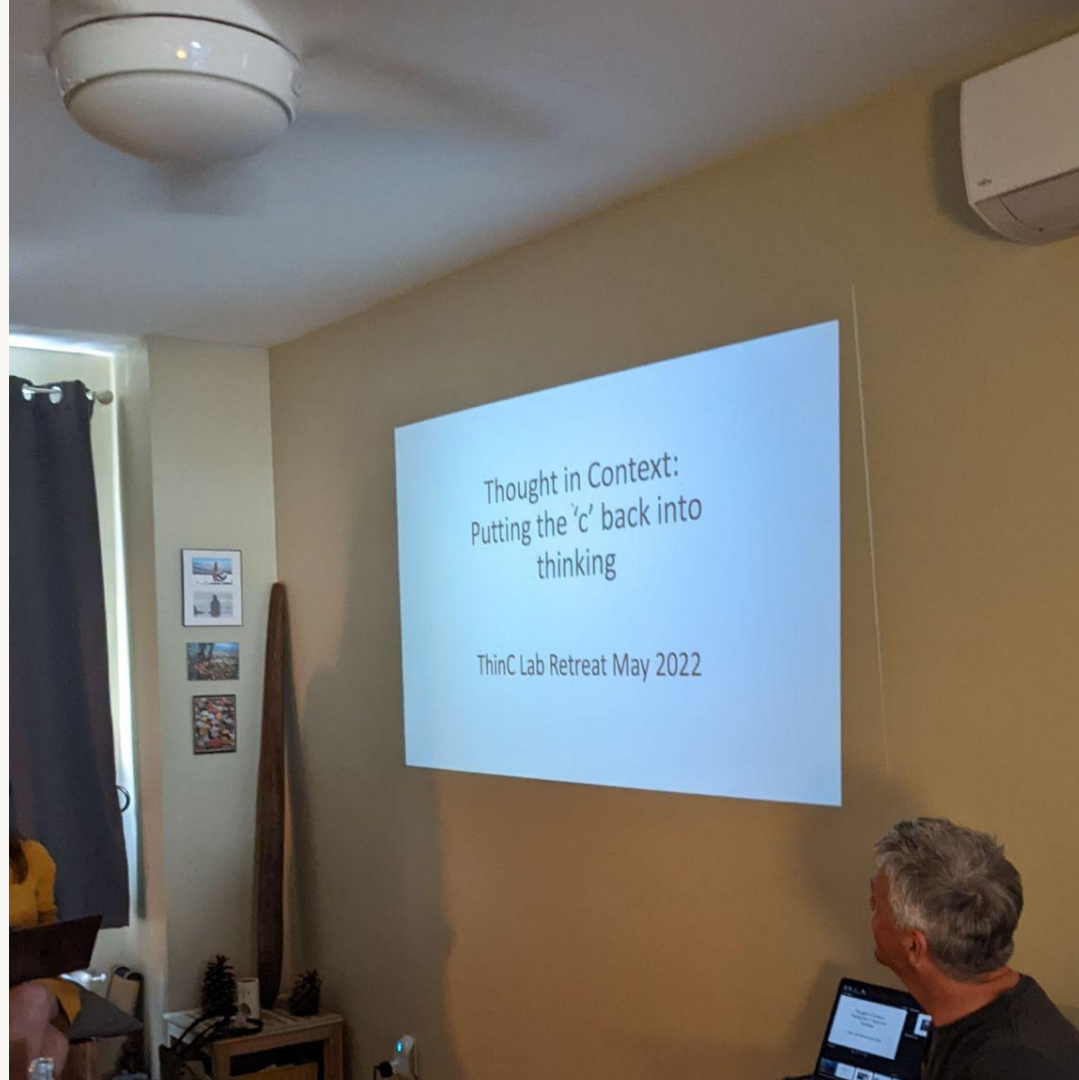


~~minGPT~~ nanoGPT



<https://github.com/karpathy/nanoGPT>

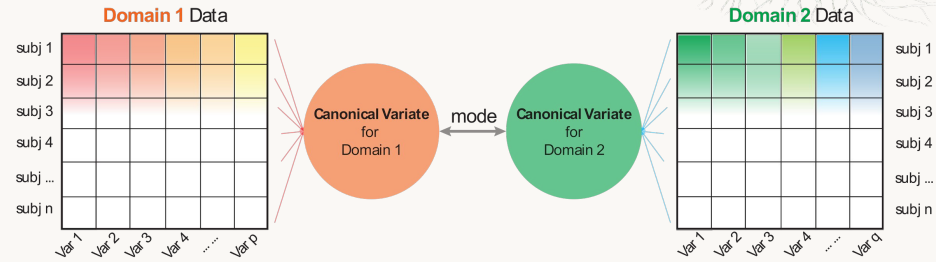
brainhack-vandy.github.io



Smallwood, ..., **Wang**, ... et al. (2021)
iScience



Brain-thought association through canonical correlation analysis

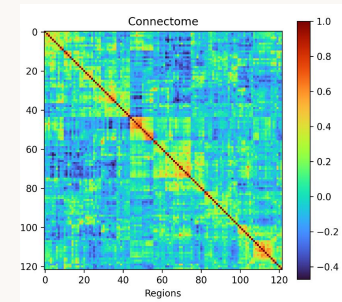


Wang et al. (2018) Psychological Science

Wang et al. (2018) NeuroImage

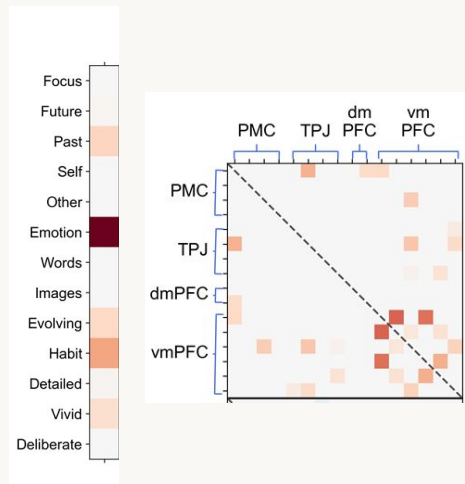
Wang et al. (2020) Scientific Reports

Wang et al. (2020) NeuroImage

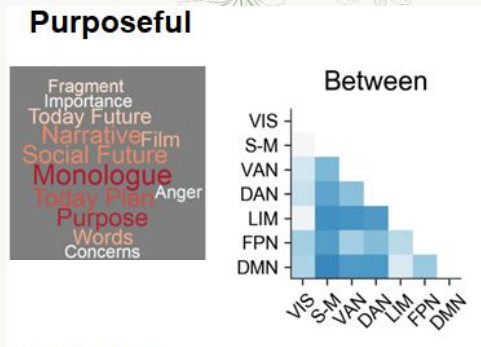
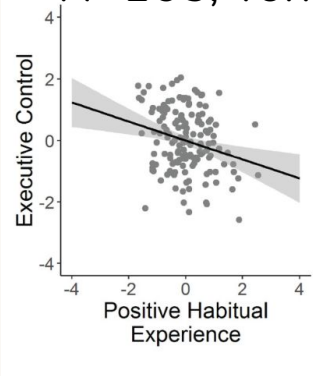


Brain-thought patterns correlate with cognitive and behavioural traits

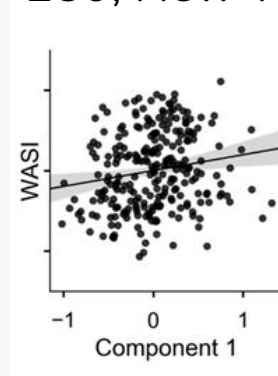
Wang et al. (2018) Psychological Science
Wang et al. (2018) NeuroImage



N=165, York



N=250, New York



Popularity of naturalistic imaging in recent years

NeuroImage 2022; special issue cover
Creative Direction & Production: Tamara
Vanderwal

3D Design: Tobias S. Hoffmann



Naturalistic neuroimaging in CNeuroMod dataset

CNeuroMod poster, CCN 2025

Credit: B. Pinsard

		Stimuli modalities						Responses				Imaging			Total	
Primary Domain	Datasets (in preparation)	Vision	Text	Speech	Audio	Music	Emotion	Memory	Physiology	Eyetracking	Actions	Speech	MRI (h/sub.)	Other (h)	# Subjects	MRI (h)
Vision	Movie10	👁️		🗣️	👂	🎵	😊		💓				10		6	60
	Friends-s01	👁️		🗣️	👂	🎵	😊		💓				10		6	60
	Friends-s02	👁️		🗣️	👂	🎵	😊		💓				10		6	60
	Friends-s03	👁️		🗣️	👂	🎵	😊		💓				10		6	60
	Friends-s04	👁️		🗣️	👂	🎵	😊		💓				10		5	50
	Friends-s05	👁️		🗣️	👂	🎵	😊		💓	👁️👂			10		5	50
	Friends-s06	👁️		🗣️	👂	🎵	😊		💓	👁️👂			10		5	50
	Friends-s07	👁️		🗣️	👂	🎵	😊		💓	👁️👂			10		5	50
	OOD Aliopauts	👁️		🗣️	👂	🎵	😊		💓	👁️👂			2		4	8
	things	👁️							📁	💓	👁️👂	👤	18		4	72
Audition	retinotopy	👁️								👁️👂	👤	1		4	4	
	floc	👁️							💓	👁️👂	👤	1		4	4	
	emotion-videos	👁️			🎵	🎵	😊		💓	👁️👂	👤	5		5	25	
	mute-music	👁️			👂			📁	💓	👁️👂	👤	4		5	20	
Language	narratives		📄	🗣️		🎵		📁	💓	👁️👂	👤	4		5	20	
	harrypotter		📄	🗣️		🎵			💓			1		5	5	
	petit-prince			🗣️		🎵			💓	👁️👂		3		5	15	
	triplets		📄	🗣️					💓	👁️👂		7		4	28	
Memory	langlocalizer		📄	🗣️					💓	👁️👂		1.5		4	6	
	multfs	👁️						📁	💓	👁️👂	👤	8		5	40	
Action	shinobi	👁️			👂	🎵			💓	👁️👂	👤	10		4	40	
	mario	👁️			👂	🎵			💓	👁️👂	👤	18		5	90	
	mariostars	👁️			👂	🎵			💓	👁️👂	👤	10		5	50	
	mario3	👁️			👂	🎵			💓	👁️👂	👤	10		5	50	
Other	hcaprt	👁️	📄	🗣️	👂		😊	📁			👤		10		6	60
	gamepad		📄								👤		1		4	4
	anat												17.5		6	
	mario_eeg	👁️			👂	🎵					👤			10	3	0
	friends_fix	👁️		🗣️	👂	🎵	😊			👁️👂			1.3		3	3.9
	movie10_fix	👁️		🗣️	👂	🎵	😊			👁️👂			1.3		2	2.6
	shinobi_training										👤			8	5	
	audiology tests													11	6	
Total	32	22	4	15	18	20	13	5	21	18	11	1	214.6	29	6	987.5
	h/modality/subject	176	20	103	147	146	100	44	173	122	86	4				

Common framework for model evaluation: Examples from the Algonauts-CNeuroMod competition



BRAINHACK
VANDERBILT
2026

ALGONAUTS
—PROJECT—



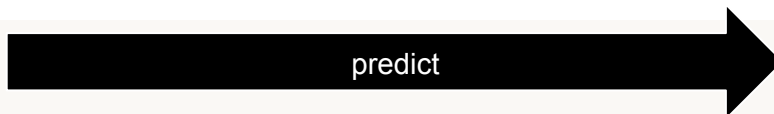
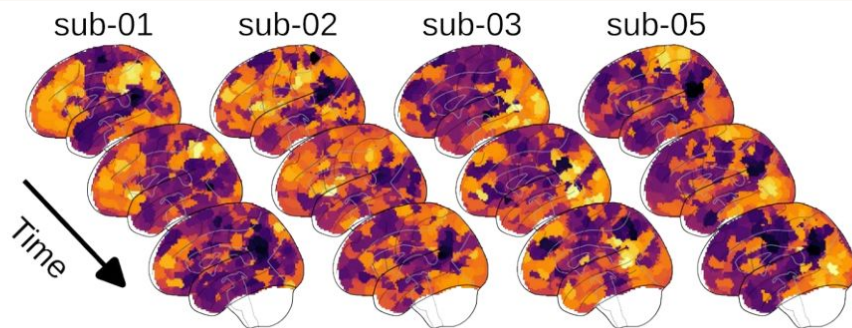
Visual frames



Audio samples

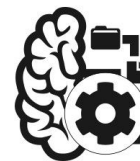


Language transcripts



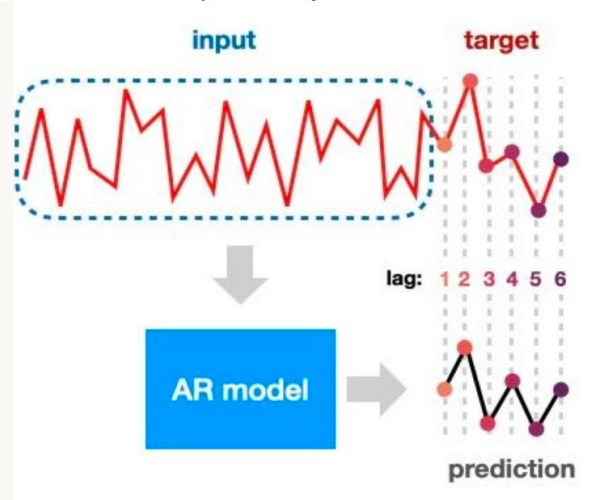
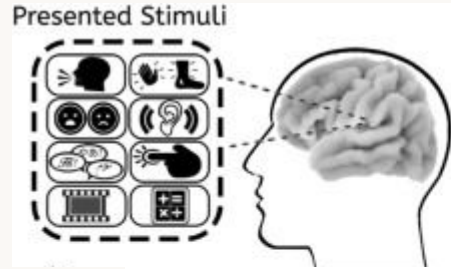
This competition will be used to generate a family of high-quality pretrained vision models robustly aligned with brain data. Gifford et al. Arxiv 2025.

giga-connectome

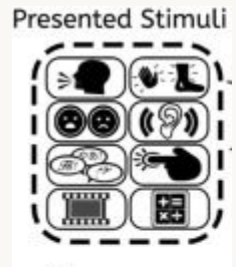


brainhack-vandy.github.io

fMRI decoding of naturalistic stimuli as a common benchmark



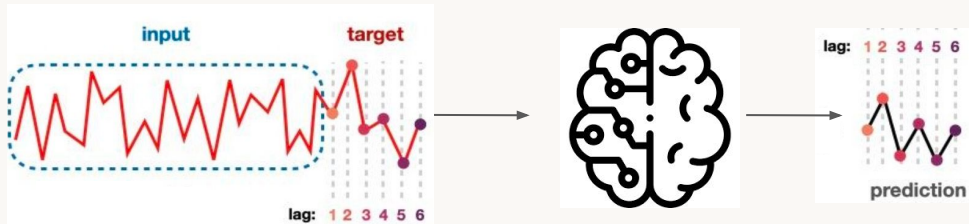
Decode



Study brain activity in daily life through experience sampling



Constraints



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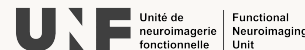
Nicolas Farrugia



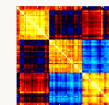
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neuroinformatics, psychiatry, neuroimaging

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